

Rockborne

Logistic Regression

Predict a probability, then decide a class

Objectives



What and why

The sigmoid and predicting a probability.



Interpreting the model

Coefficients as odds ratios.



Evaluating and tuning

ROC, AUC and the decision threshold.

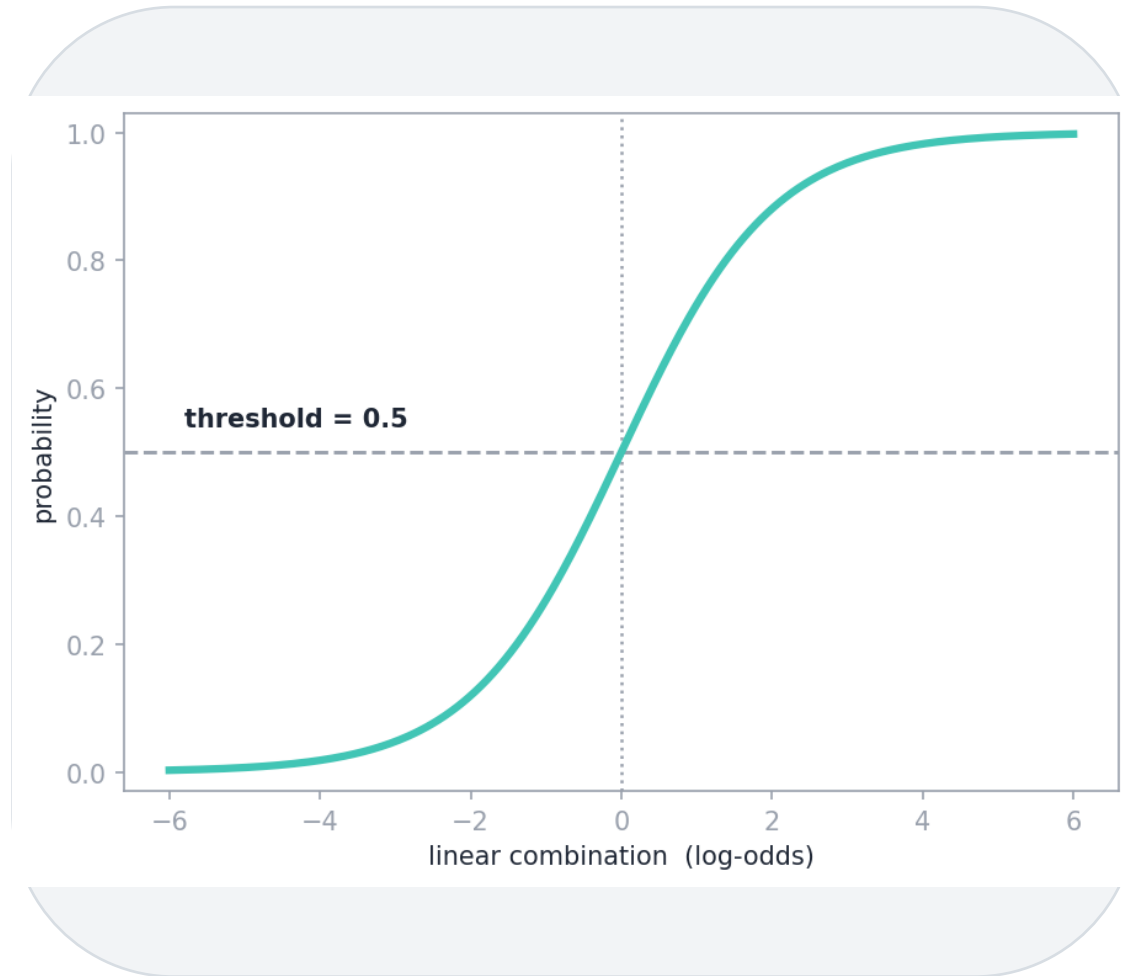
Introduction



What is Logistic Regression?

What is Logistic Regression?

- A classification method that predicts a probability between 0 and 1.
- The sigmoid squashes a linear combination of features into that probability.
- A threshold then turns the probability into a class.
- The interpretable baseline, and the classic credit scorecard.



Linear vs Logistic Regression

Same linear core, wrapped in a sigmoid so the output is a probability.

Linear regression

- Predicts any real number.
- Fits a straight line.
- Used for regression.

Logistic regression

- Predicts a probability (0 to 1).
- Fits an S-shaped sigmoid.
- Used for classification.

Interpreting the Model: Odds Ratios



1

Each coefficient is a change in the log-odds.

2

$\exp(\text{coefficient})$ gives the odds ratio.

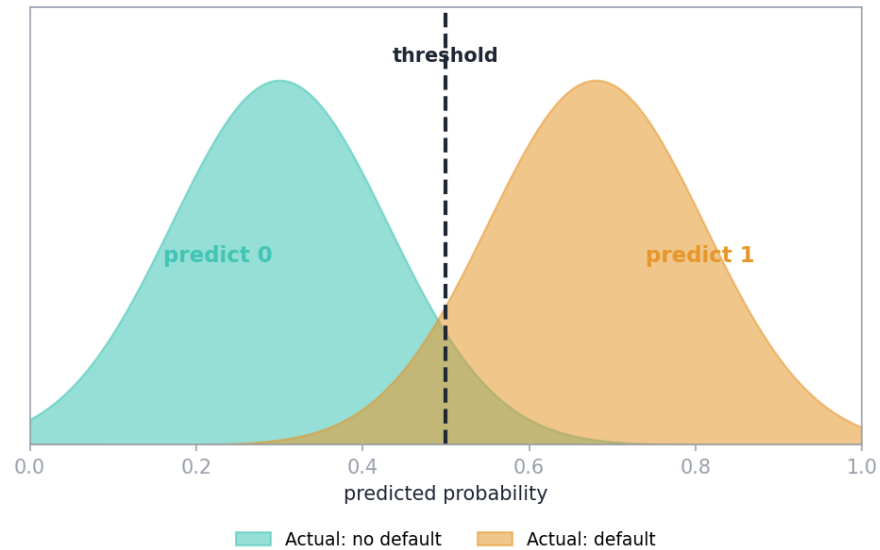
3

Odds ratio > 1 raises the odds; < 1 lowers them.

4

This is what makes logistic regression explainable.

From Probability to Decision



The model outputs a probability.

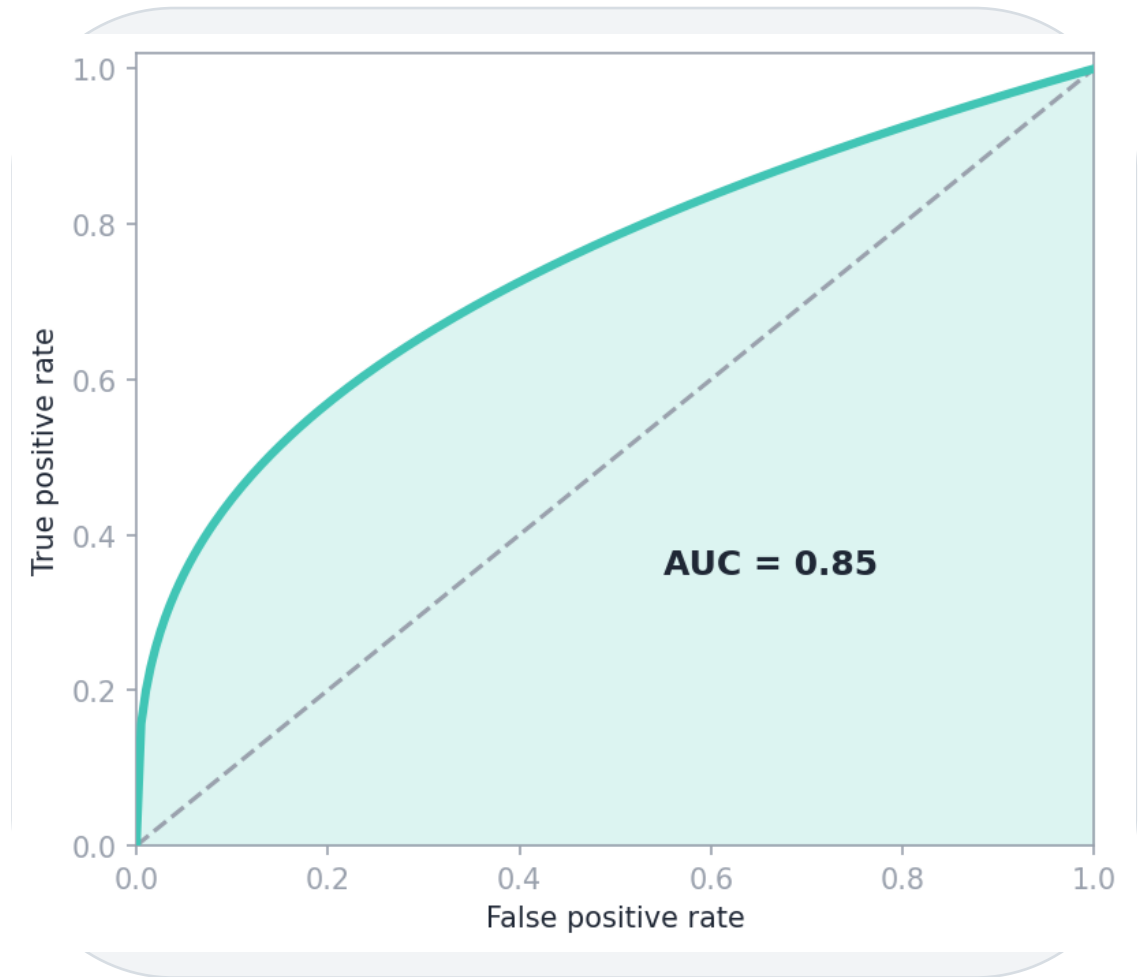
A threshold (default 0.5) turns it into a class.

Lower the threshold to catch more positives (higher recall)...

...at the cost of more false alarms. Set it to the business cost.

Evaluating: ROC and AUC

- The ROC curve plots the true-positive rate against the false-positive rate across thresholds.
- AUC (area under the curve) summarises performance in one number.
- AUC = 1 is perfect; 0.5 is no better than chance.
- Great for comparing models and for imbalanced data.



Evaluating: the Classification Report



Accuracy

Correct predictions over all predictions.



Precision

Of those flagged positive, how many really are.



Recall

Of the real positives, how many we caught.



F1-score

The harmonic mean of precision and recall.

Regularisation

C controls the strength: a smaller C means stronger regularisation.

L1 (Lasso)

- Can shrink some coefficients to exactly zero.
- Performs automatic feature selection.
- Useful with many, sparse features.

L2 (Ridge)

- Shrinks coefficients smoothly toward zero.
- Keeps all features (the default).
- Stabilises correlated features.

Pros and Cons

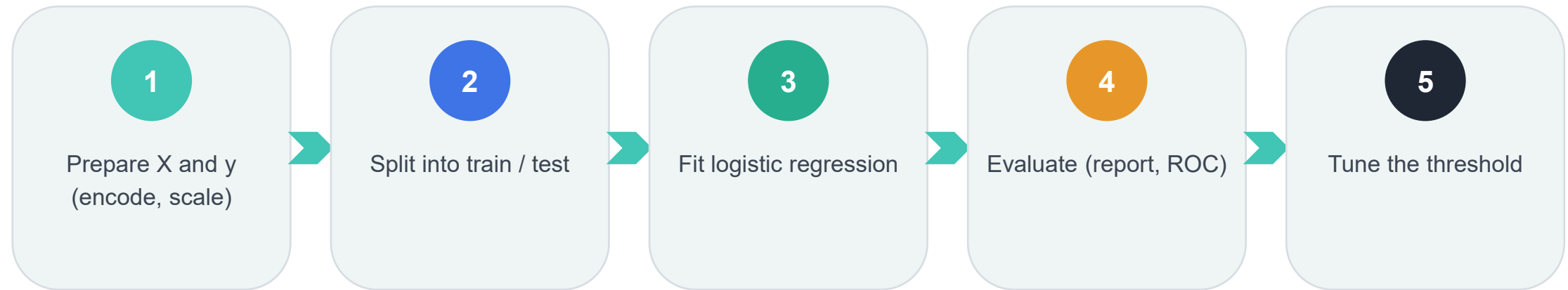
Pros

- Interpretable (odds ratios).
- Outputs calibrated probabilities.
- Fast to train; a strong baseline.

Cons

- Assumes a linear relationship in the log-odds.
- Needs encoding and feature scaling.
- Struggles with complex, non-linear boundaries.

Methodology



In this lesson, we explored



What and why

Sigmoid, probability, and the decision threshold.



Interpretation

Coefficients as odds ratios; an explainable model.



Evaluation

Report, ROC/AUC, threshold tuning, regularisation.

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Time to get hands-on in Python

We move from theory to practice with a guided walkthrough you run yourself in Python.



Logistic Regression - Guided Walkthrough



Then: Independent Classification Labs

Digging Deeper?



[LogisticRegression \(API\)](#)

scikit-learn



[Logistic Regression \(User Guide\)](#)

scikit-learn



[Logistic Regression](#)

Wikipedia

Questions?